

EduConnect Digital Solutions

Full Product Feature Specification

Post-MVP — All Dashboards — Version 2.0 | 2026

Features are grouped by dashboard and ordered by build priority.

1 — Admin Dashboard: Missing Core Features

► **CRITICAL** Fee Management & Invoicing

No fee system currently exists. During my research, I realized that most SISs have an inbuilt payment system integrated with local Payment Systems e.g (Remita, Flutterwave, Paystack). The school collects money, it needs to track who has paid, who hasn't, and how much is owed.

- Fee structure configuration: define fee types (tuition, development levy, PTA, uniform, etc.) per class per term if recurring.
- Bulk invoice generation: one action generates invoices for all students in a class or the whole (OPTIONAL)
- Per-student invoice: auto-populated with fee items based on the student's class
- Payment recording: manual entry of payment received (cash, bank transfer, POS) (If integrated with Remita, this is done automatically)
- Automatic EduConnect or School headed receipt generation on payment confirmation, downloadable as PDF.
- Outstanding balance tracker: red/amber/green status per student
- Overdue alerts: admin notified when term is X days old and Y students still have unpaid balances

GAP ► **HIGH** Attendance Overview & Management

Admin currently has no visibility into school-wide attendance. They cannot see which classes had poor attendance today, which students are chronically absent, or what the school-wide attendance rate is.

- School-wide daily attendance summary: % present by class, updated as teachers mark and submit.
- Absentee list: every student marked absent today, exportable to PDF for follow-up
- Chronic absentee flag: students absent more than X% of school days this term (configurable threshold, predefined by Admin)
- Class-level attendance report: attendance rate per class per week/month/term
- Individual student attendance history: full calendar or tabled view of present, absent, late.
- Bulk attendance import for schools that collect paper registers and enter later

► **HIGH** Results & Academic Performance Overview

Admin cannot currently see how the school is performing academically. No aggregate grades view, no class averages, no subject pass rates.

- Term results overview: average score per class per subject once grades are published
- Subject pass rate: % of students who passed each subject, per class
- Class performance ranking: which class had the highest average performance this term
- Individual student academic standing: current term grades at a glance

- Grade submission status: which teachers have published grades, which have not (critical for term-end processing)
- Comparative view: this term vs last term performance by class and subject

► **HIGH Report Card Generation**

Report cards were listed as a premium feature on our website but are not yet built. Without this, we cannot replace most school's existing systems.

- Trigger: admin initiates report card generation for a class or the entire school once grades are finalized by the teacher.
- Template: school logo, school name, student name, class, term, academic year, table of subjects with CA scores, exam scores, total, grade, position, remarks
- Class position and subject position auto-computed from entered grades
- Teacher's remarks field: free text entry by class teacher before cards are finalised
- Head Teacher / Principal's remarks field
- Grade key printed on card (A1 – Excellent, F9 – Fail, etc.)
- Bulk PDF generation: all report cards for a class in one download, one per student
- Individual student report card PDF downloadable by parent from their portal
- Lock mechanism: once published, report cards cannot be edited without admin override.
- FR-DEV NOTE: The report card template must be configurable by the school (logo, colour) but must also work out of the box with zero configuration for schools that just want to get it done. Provide two or three template options, do not make template customisation the first thing the admin encounters.

► **MEDIUM Timetable Management**

Schools need to create and manage their weekly timetable within EduConnect. Without this, the platform is disconnected from the school's daily operations.

- Timetable view for teachers: each teacher sees only their own timetable
- Timetable view for admin: full school timetable with filter by class or teacher
- Export to PDF for printing and displaying on noticeboards

► **MEDIUM School-Wide Communication & Announcements**

Communication from school to parents and staff needs to be a managed function within EduConnect.

- Announcement composer: admin writes a message, selects audience (all parents, all teachers, specific class, specific group, specific individual), and sends
- Announcement delivery channels: in-app/web notification, email, SMS (where configured)
- Announcement history: archive of all past communications

2 — Teacher Dashboard: Missing Core Features

► **CRITICAL** Attendance Marking

This is the single most commonly used teacher action in any SIS.

- Daily class register: teacher opens their class for today and sees the full student list
- One-tap marking: tap student name to toggle Present / Absent / Late
- Bulk mark all present: one button to mark everyone present, then individually mark exceptions (faster for most days)
- Latecomers: marking a student as Late prompts for an optional time entry
- Absence reason: optional field maybe dropdown to record why a student is absent (sick, family, late, "type in", unknown)
- Save & submit: once marked, attendance is locked for that day (editable by admin only with audit trail)
- Retroactive entry: teacher can submit attendance for a missed day with a note explaining the delay (OPTIONAL, backend modifications)
- FR-DEV NOTE: a mark attendance button should be in the default landing page for the teacher dashboard on page load.

► **HIGH** Teacher's Own Timetable View

Teachers currently have no way to see their schedule within EduConnect. This is basic.

- Personal timetable: the teacher sees their own weekly schedule — which class, which subject, which period
- Today view: at a convenient position in the dashboard, show today's lessons in chronological order
- Link from timetable slot to the class: tap a lesson to go directly to that class's attendance or gradebook (Semi Implemented)

► **HIGH** Teacher-to-Admin & Teacher-to-Teacher Communication

Teachers have no way to communicate within EduConnect. They are probably using WhatsApp for everything school-related, which is unprofessional and unaudited.

- Internal message system: teacher can send a message to admin, a colleague, or a department
- Announcement inbox: teacher receives school-wide and class-specific announcements from admin in one feed.

► **MEDIUM** Student Performance Snapshot (Per-Teacher View)

A teacher should be able to see how their students are performing at a glance, not only when entering grades.

- Subject ranking: top 5 and bottom 5 students in each subject the teacher teaches
- Comparison to class average: how does each student sit relative to the class mean

3 — Parent Dashboard: Missing Core Features

► CRITICAL Real-Time Attendance Visibility

- Attendance calendar: full month view showing each school day colour-coded (green = present, red = absent, yellow = late)
- Today's attendance status: prominently displayed on the parent dashboard home — 'Amina was marked Present today at 8:02am'
- Attendance statistics: current term attendance rate as a percentage, with a visual indicator
- Absence alert: push notification sent to parent the moment the attendance is concluded
- Weekly summary: every Friday, parent receives a summary of the week's attendance

► CRITICAL Grade & Academic Results Visibility

Parents can see a 'child information overview' but not actual grades. This is the second most important thing a parent wants.

- Current term grades: parent sees each subject with scores entered so far (CA1, CA2, Exam as they are released by school)
- Term result: once report cards are published, parent sees the full report card in the portal
- Subject performance breakdown: score per subject with grade letter and remark (Excellent, Good, Pass, Fail)
- Historical results: previous terms' grades accessible in a timeline
- Class position: where does my child rank in their class this term
- Report card download: parent can download a PDF of the official report card once published

► HIGH Fee Payment Status

Parents need to see their fee obligations clearly. Currently there is no fee information in the parent portal.

- Outstanding balance: clearly displayed at the top of the parent portal — 'You owe ₦50,000 for Second Term 2026'
- Fee breakdown: what the amount is made up of (tuition, levies, etc.)
- Payment history: all payments made, with dates, amounts, and downloadable receipts, again, Educonnect of School based.
- Payment deadline: how many days until the next fee deadline.
- Fee overdue alert: push notification when a payment is past due

► HIGH Parent-to-School Communication

Parents currently have no way to contact the school or their child's teacher through EduConnect. They resort to personal phone numbers.

- Message to admin: parent can send a message to the school administration
- Response time indicator: 'The school typically responds within 1 business day' (configurable by admin)
- Message templates: pre-written message starters for common queries (grade query, absence explanation, fee query, meeting request)

► MEDIUM School Calendar & Events Visibility

Parents should be able to see the school's academic calendar, upcoming events, and important dates.

- Term dates: start and end dates for each term, visible to parents
- Upcoming events: sports days, visiting day, open day, parent-teacher meetings, school celebrations
- Exam timetable: when published by school, parent can see the exam schedule for their child's class
- Fee deadlines: payment due dates visible on the calendar
- Holiday dates: school holidays and breaks

4 — Cross-Platform UX & Design Standards

► **CRITICAL** Mobile-First Design for All Three Dashboards

The primary device for teachers, parents, and many admins in Northern Nigeria is a low-to-mid-range smartphone.

- All three dashboards must function completely and comfortably on a 5-inch mobile screen
- Navigation must be bottom-bar or hamburger-menu based, not top-nav sidebar
- Forms must use large input fields with appropriate mobile keyboards (numeric keypad for scores, phone keyboard for phone numbers)
- Tables must scroll horizontally on mobile and never force horizontal page scroll. Or completely redesign tables for mobile for full one time view.
- No hover-dependent interactions, everything must work on touch only
- Loading states must be explicit (**IMPORTANT**), skeleton loaders or spinners so the user knows the app is working on a slow connection
- DESIGN SYSTEM NOTE: Build a single component library that is mobile-first and apply it consistently across all three dashboards. Inconsistency between dashboards (different button styles, different navigation patterns) is a sign of an unfinished product.

► **HIGH** Onboarding Flow for New Schools

There is currently no structured onboarding flow. A new school logs in and sees a blank dashboard with no guidance on what to do next.

- Step-by-step setup wizard triggered on first login: 1) Complete school KYC profile, 2) Set up academic year and terms, 3) Create classes and subjects, 4) Add staff, 5) Import or add students. Currently Subjects are hardcoded on the frontend, this should be a backend affair.
- Progress indicator: shows how many setup steps are complete and which are remaining
- 'Skip for now' option on each step for users who want to explore before completing setup
- Completion celebration: once all setup steps are done, a clear confirmation screen with 'Your school/account is ready. Here's what to do next.'
- Contextual empty states: when a section has no data yet, show a friendly illustration and a clear call-to-action.
- Help tooltips: on first visit to any new section, a brief tooltip explains what it is and what to do

► **CRITICAL** Consistent Loading, Error, and Empty States

- Loading states: every data-fetch operation must show a loading indicator. Never show a blank screen.
- Error states: if a network request fails, show a human-readable message ('Could not load your class list. Check your connection and try again.') with a retry button, never show a raw error code or a blank page
- Empty states: every list, table, or data view must have a designed empty state with an illustration, an explanation, and a clear next action. Empty states are an onboarding opportunity, not an error.
- Success states: important actions (attendance submitted, grades saved, message sent) must give explicit positive confirmation. A toast notification or a checkmark animation.

► **MEDIUM** Notification System Architecture

EduConnect sends notifications but the system for managing them is undefined. This needs to be

designed before building the advanced notification features in Part 4.

- Notification types: in-app (bell icon), push notification (mobile/PWA), email, SMS
- Per-user notification preferences: user can choose which notification types they receive and via which channel
- Unread count badge on the notification bell icon
- System notifications vs school notifications: distinguish between system messages (password reset, security alert) and school-generated messages (announcement, attendance alert)

Part 5 — Full Feature Roadmap & Build Priority

The table below sequences every feature. Gap fixes and differentiators into a single prioritised build order. 'Now' means this sprint cycle (core product completion). 'Phase 2' means 3–6 months (depth and differentiation). 'Phase 3' means Year 2 once the core product is stable and revenue is growing.

Feature	Build When	Rationale
Academic Structure Setup (Classes, Subjects, Arms)	Now	Without this, nothing else works. First-time setup wizard. Must be built before any school goes live.
Academic Calendar & Three-Term Structure	Now	Required for term-contextual grades and attendance. Foundation for all time-based features.
Attendance Marking; Teacher Dashboard	Now	The most frequently used teacher action.
Report Card Generation	Now	Every school issues report cards every term.
Fee Management & Invoicing	Now	Schools collect money. They need to track it. Non-negotiable for any paying client.
Attendance Visibility. Parent Dashboard	Now	Parents opening the platform want to know: is my child at school? This must be the first thing they see.
Real-Time Attendance Alerts (E2)	Now	Parents notified the moment child is marked absent. Delivers on the platform's core promise.
Grade Visibility. Parent Dashboard	Now	Parents need to see their child's actual scores, not just that the school has activity.
Mobile-First Responsive Design (all 3 dashboards)	Now	Desktop-only design might kill adoption in the target market.
Onboarding Wizard for New Schools	Now	Without guided setup, schools might abandon the platform on initial load.
Loading, Error & Empty States (all screens)	Now	Blank screens and raw error codes lessen trust. Define and build all three states for every screen.
Nigerian Grading Framework A1–F9	Now	Must be correct before the first school uses the gradebook. A1-F9, WAEC standard, pre-loaded.
Nigerian Academic Calendar	Now	Three-term structure baked in. Schools should not have to configure this from scratch.
School Term Intelligence Report	Now	Auto-generated end-of-term PDF report. Impressive in demos and pitches. High perceived value.
Fee Status Visibility - Parent Dashboard	Phase 2	Critical for reducing admin follow-up calls and possible wait time.
Timetable Management	Phase 2	Needed before targeting secondary schools that have complex period structures.

School-Wide Communication & Announcements	Phase 2	Currently parents receive notifications but from no structured source.
Teacher Timetable View	Phase 2	Basic feature every teacher expects. Build when timetable management is done.
Teacher-to-Admin Messaging	Phase 2	Replaces WhatsApp for internal school communication. Audited and professional.
Parent-to-School Messaging	Phase 2	Replaces personal phone numbers for parent queries. Critical for parent retention.
Admin Attendance Overview Dashboard	Phase 2	School-wide attendance at a glance.
Notification System Architecture	Phase 2	Define the full notification framework before building more notification types.
Student Performance Overview - Admin	Phase 2	Aggregate academic performance view. Needed before any analytics can be meaningful.
Offline Mode for Attendance & Grades	Phase 2	Highest-impact technical differentiator. Build once the online flow is stable and tested.
WhatsApp Notification Delivery	Phase 2	Requires WhatsApp Business API. Single highest-impact parent engagement feature.
Multi Language Interface	Phase 2	Strategic differentiator. Must use professional translation, not machine translation.
Assignment & Homework Manager	Phase 2	Links teacher workflow to parent visibility. Parents see homework deadlines.
Behaviour & Conduct Log	Phase 2	Completes the student record.
Custom Roles & Permissions	Phase 3	Required before targeting larger schools with Head of Department, Bursar, Form Master roles.
PWA with Partial Offline Support	Phase 3	Significant engineering investment. Build once the core online product is proven.
Full At-Risk Prediction Engine with Intervention Log	Phase 3	ML-based prediction requires 2+ years of data from 500+ students to be meaningful.
Multi-Campus / School Group Management (C4)	Phase 3	Major architecture work. Entices NGO and school group market.
EduConnect for NGOs Programme Dashboard	Phase 3	The institutional product. Requires multi-school data at scale.
School Public Noticeboard Page (F2)	Phase 3	Marketing value. Build when engineering bandwidth permits.